

Reducing supply risk of critical materials for clean energy via foreign direct investment

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Existing research on the security of the supply of critical materials for clean energy generally aggregates information at the country level, a practice that obscures the extensive role of foreign direct investment (FDI) in the production of critical materials. FDI refers to an ownership stake in a company or project by an overseas investor. Here we establish a database for global mining of lithium, cobalt, nickel and platinum at company level, covering 240 countries and regions. We show that 47% of lithium, 71% of cobalt, 41% of nickel and 34% of platinum mined in 2019 were under FDI. We then explore how FDI may affect supply risks by proposing a supply risk index that allocates production of the critical materials to the country of origin of investors instead of the country where production is located. We present upper and lower bounds of the supply risk index that reflect scenarios where either all investors or only state investors prioritize the home-country demand, respectively. This study presents an approach for assessing the national supply risks of critical materials, considering the geographical allocation of FDI.

Clean energy technology deployment is rapidly expanding to address urgent climate challenges¹. Compared with conventional energy technologies, clean energy applications require a wider range of materials, many of which are labelled ‘critical materials’^{2–5}; for example, lithium, cobalt, nickel and graphite for lithium-ion batteries^{6,7}, platinum group metals for electrolyzers and fuel cells⁸, and rare earth metals for permanent magnet motors⁹. Ensuring stable supplies of critical raw materials has become a prerequisite for sustainable energy transition^{10–13}.

These critical materials, however, often show significantly unbalanced production and consumption patterns. Their primary supply mostly originates from regions such as South America, Africa and Oceania¹⁴, while their demand mainly concentrates in major economies such as the USA, the European Union (EU) and China.

This makes consuming countries highly import-dependent and thus face challenges of potential materials supply disruption¹⁵. These risks have been widely discussed by studies using the methods of material flow analysis^{9,16–19}, criticality assessment^{8,20–23} and complex network analysis^{6,24,25}.

Nevertheless, many studies that concentrate on import dependence overlook the heterogeneity and concentration of supply sources, thus failing to fully capture the distribution of risks. Moreover, previous studies generally centre on the national level and rarely address the company level. The usual focus overlooks that major economies are often home to multinational mining companies that have a certain proportion of property rights in overseas mines through foreign direct investment (FDI)²⁶. FDI may alleviate supply shortage

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concerns of general material consumers of the home country through various channels. Examples include a proclivity to exploit enhanced downstream capacity in the home country²⁷, following of policy incentives that serve the home country's resource security (such as those in the US Inflation Reduction Act and EU Critical Raw Materials Act)²⁸, and favoring economic and national security interests of home-country investors, a channel that generally applies to state-owned companies²⁹.

Previous studies have shown the role of FDI in reducing raw material supply risks for some specific countries such as China and Japan^{30–33}. Nevertheless, a comprehensive overview of FDI in global critical material markets is still not available. Note that while a country's FDI (for example, US investment) could reduce its domestic supply risk, it may also limit the raw material availability for other countries (for example, China and EU). Thus, it is vital to track FDI from a global perspective and systematically consider its impact on national supply risks.

Here we aim to address these gaps by mapping the global distribution of FDI and assessing its potential role in supply risk mitigation in the case of four critical materials: lithium, cobalt, nickel and platinum. These materials are selected because they are deemed critical for low-carbon mobility and power transition^{34,35} while considering the availability of required data and our research expertise^{8,19,36–39}. We first establish a company-specific global production database for these materials, including mine locations, operators, shareholders and corresponding FDI ownership hierarchy, and country affiliation of relevant companies. Then we trace the global production and trade network of the four materials for all countries and their dependent territories (240 regions in total) in 2019 when the latest data are available. The tracking focuses on the mining stage, while trade of lithium carbonate and platinum refined metals is also included given their common vertical integration of mining and refining processes. The two perspectives on production and international trade are thereafter presented: one by geographical location and the other by country of origin of the investors, which assumes that production control is proportional to the shareholders' ownership of equity. We further present a supply risk index (SRI) to quantify geopolitical supply risk faced by each country. Specifically, we derive an 'original SRI' and an 'adjusted SRI', respectively utilizing the two abovementioned perspectives. Our SRI considers both import dependency and supply concentration, assuming that importing is riskier than domestic production (this assumption necessitates further analysis as diversified imports could be more secure than concentrated domestic production).

We reveal significant differences between the geographical and company perspective of global production and trade networks of the four materials due to extensive FDI. We find that in the scenario whereby all investors follow the 'home-country priority' mode, FDI represents great potential for mitigating geopolitical supply risk for the USA's lithium, the United Kingdom's platinum, Japan's nickel, and China's lithium and cobalt. We also consider another scenario where only state investors prioritize home-country interests, reflecting their stronger ties to national interests compared with private investors. Here, the benefits of FDI are primarily evident in cobalt and nickel, dominated by Chinese state-owned enterprises. The comparison of the original SRI and the adjusted SRI represents the potential role of FDI in supply risk mitigation, with the reality situated somewhere between the two proposed scenarios. Our results highlight the importance of considering FDI in critical material supply security assessments.

Results

Global material production control database

Figure 1 shows the production control hierarchy of global FDI-related mines, and Supplementary Figs. 1–4 and Tables 1–4 contain further details. The location of a company's headquarters is defined as its country affiliation. The production 'control' is quantified on the basis of the company ownership of mine projects and subsidiaries (see Methods for

details). In 2019, American companies controlled the largest amount of lithium mines overseas with a total production of 21 kt, mainly from Albemarle's Atacama mine in Chile, and its 49% ownership of Talison and FMC's Hombre Muerto mine in Argentina. Chinese companies came in second, controlling production of 15 kt in Australia and 3 kt in Chile, mainly from Tianqi's 51% ownership of Talison and 26% ownership of SQM, and Ganfeng's 43% ownership of Reed Industrial Minerals. These lithium companies, which engaged in both mining and refining operations, direct their overseas investments in minerals to primarily meet the demand for refining capacity in their respective home countries. Glencore, a Swiss company, controlled the two largest cobalt production mines in the Democratic Republic of the Congo (DRC), while Chinese companies came in second with 22 kt in total, mainly from China Molybdenum's 56% ownership of the Tenke Fungurume mine in DRC and Huayou Cobalt's Luiswishi mine in DRC. Chinese companies also controlled the largest overseas nickel supply, totalling 287 kt, mainly from Jinchuan and Tsingshan. Brazilian company Vale controlled the second-largest overseas nickel supply of 187 kt. British companies controlled most of overseas platinum production of 42 tons in total, which was mainly contributed by Anglo American Platinum. South African companies were the second-largest overseas platinum supply owner with 14 tons in total, mainly from Impala Platinum's Zimplats mine and Sibanye–Stillwater's Stillwater & East Boulder mine in the USA.

Impact of FDI on the production side

On the basis of the above company-specific database of the four materials, the national mineral production from a geographical perspective was 'reallocated' to derive the mineral production from a company ownership perspective (Fig. 1; see Methods for details). We find significant disparities between the physical geographical location of supply and supply ownership, as shown in Fig. 2 and Supplementary Tables 5–8.

Results indicate that 47% of lithium, 71% of cobalt, 43% of nickel and 34% of platinum in their corresponding global mine production are FDI-controlled in 2019. Major geographical producers of lithium were Australia, Chile, China and Argentina, with production shares of 52%, 22%, 12% and 7%, respectively. After reallocation, Chinese companies supplied the most lithium with a 33% share in total, followed by companies from Australia (26%), the USA (25%) and Chile (10%). The DRC dominated geographical cobalt production with a 69% share, while Swiss companies and Chinese companies were the top two cobalt suppliers from a company ownership perspective, with production shares of 29% and 17%, respectively. Production share of companies from the DRC was 11% after the reallocation. Indonesia, the Philippines and Russia were the top three geographical producers of nickel, with production shares of 34%, 13% and 11%, respectively. Chinese companies became the second-largest group of nickel suppliers after the reallocation, with a production share of 16%. South Africa, Russia and Zimbabwe were the top three geographical producers of platinum, with South Africa dominating due to its resource endowment. After reallocation, the production share of South African companies was 58%, remaining the largest supplier group. United Kingdom and Russian companies were the second and third largest production groups from a company ownership perspective, accounting for 23% and 13% of global platinum supply, respectively.

The Herfindahl–Hirschman Index (HHI) was utilized to quantify production concentration which ranges from 0 to 10,000, with larger numbers indicating greater concentration. HHI shows that production of lithium and platinum from both the geographic and company ownership perspectives, along with cobalt production geographically, is 'highly concentrated' (see Fig. 2 caption)⁴⁰. Mining production for all four materials is more diversified from the company ownership perspective compared with the geographical perspective. Among the four materials, cobalt production concentration has the greatest difference between the two perspectives (Table 1). These two perspectives of the HHI can serve two categories of decision making: one would be

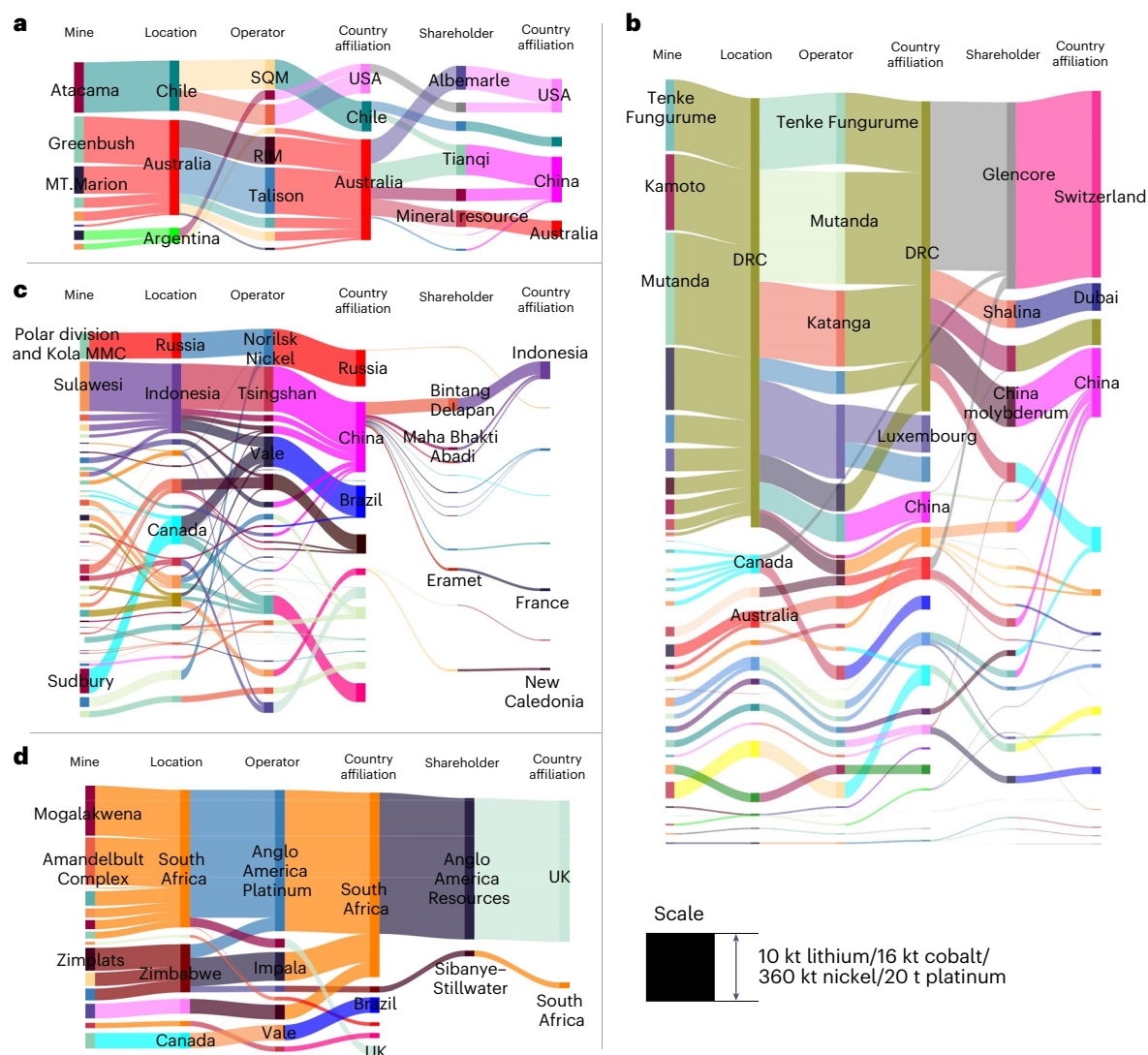


Fig. 1 | Locations, operators and shareholders of global FDI-related mines in 2019. a–d, Mine information for (a) lithium, (b) cobalt, (c) nickel and (d) platinum. Flow width is proportional to material production amount. Colours

are coded by country affiliation, which is defined by the headquarter locations of companies. For clear visualization, only the names of the top three mines, companies and countries are shown.

suitable for scenarios where decision-making processes are exclusively managed by companies, the other is appropriate in circumstances where states hold that authority.

Impact of FDI on international trade

To explore the potential impacts of FDI on the import risks faced by each country, we ‘adjusted’ the global trade flows of lithium (minerals and lithium carbonates), cobalt (ores), nickel (ores) and platinum (ores and refined metals) by treating the overseas production controlled by a country’s companies through FDI as that country’s domestic production, as shown in Fig. 3. For example, China imported 48 kt lithium from Australia and Chinese companies controlled 15 kt lithium production in Australia. Thus, the ‘adjusted trade flow’ from Australia to China would be 33 kt. If the production controlled by FDI between trade partners is larger than the trade amount, the adjusted trade amount is set as 0. Details about adjusted trade flows are presented in Supplementary Table 9 and corresponding text notes in the Supplementary Information.

The findings indicate a significant overlap between partner countries for FDI and pivotal trade flows of clean energy materials, which were mainly from Australia to China for lithium (48 kt), the DRC to China

for cobalt (75 kt), Indonesia to China for nickel (161 kt) and South Africa to the United Kingdom for platinum (14 tons). When FDI-controlled production between trade partners is subtracted, China’s imports of lithium from completely external sources are reduced by 31%, cobalt by 25% and nickel by 45%. Meanwhile, the USA’s lithium imports, Japan’s nickel imports and the United Kingdom’s platinum imports saw decreases of 88%, 58% and 35%, respectively. All other trade flows related to FDI were adjusted to zero. These trends underscore the concentration of clean energy materials trade in a handful of countries both from geographical and company ownership perspectives.

Quantifying supply risks at the country level

With a focus on the geopolitical factors, we developed the SRI to quantify the supply risk for these four critical materials faced by each country (see results of the top 10 SRIs in Fig. 4 and more details in Supplementary Figs. 9–12). The SRI comprises three sub-indicators: (1) import dependency, (2) share of national imports in total global imports and (3) likelihood of supply disruption. These sub-indicators were calculated mainly on the basis of the geographical production and international trade data of the four materials (Supplementary Figs. 5–8). The likelihood of supply disruption was calculated by further considering the

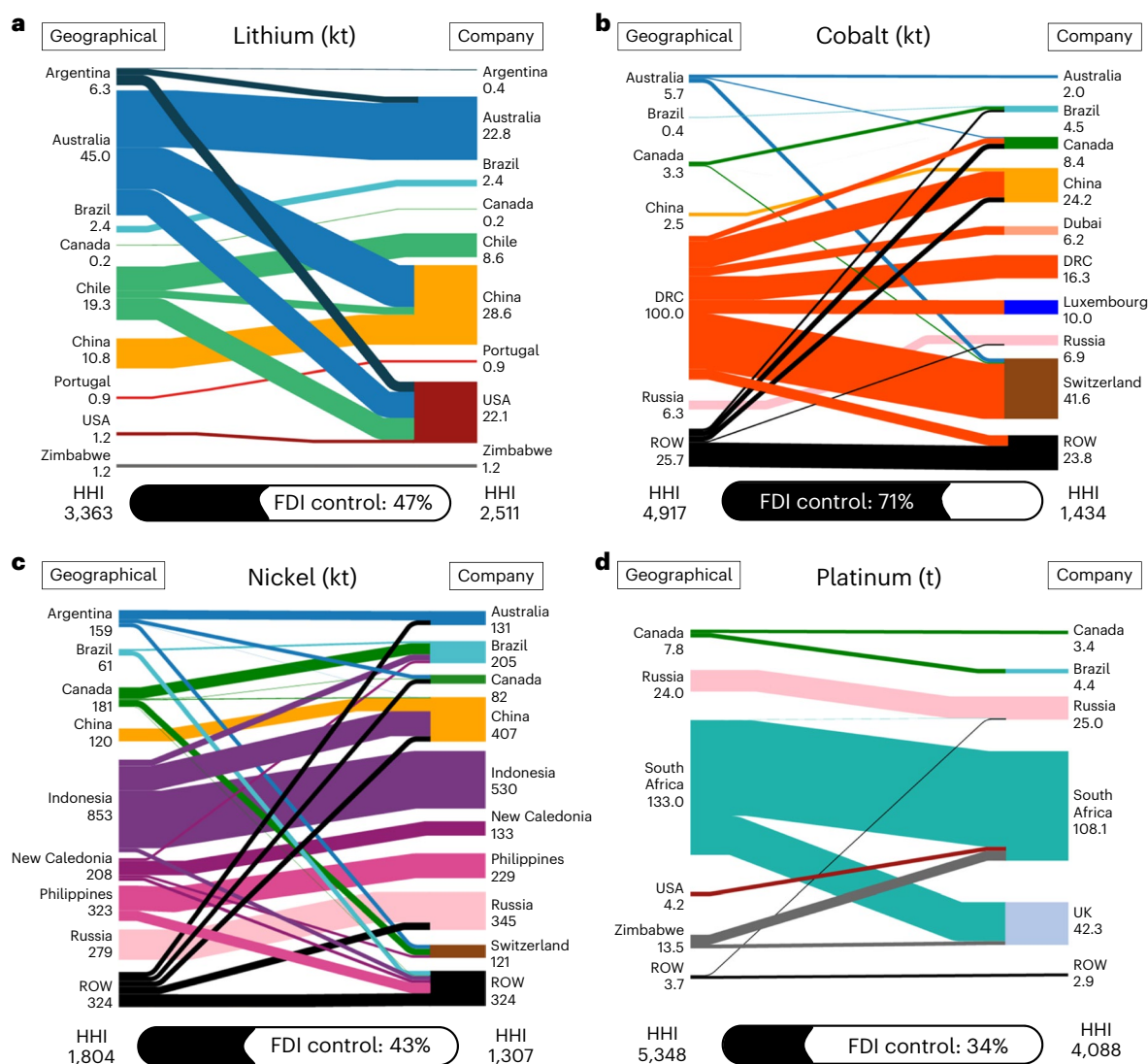


Fig. 2 | Material production from a geographical perspective (left) and a company ownership perspective (right). a–d, Mineral production of (a) lithium, (b) cobalt, (c) nickel and (d) platinum. The colours of the rectangles on both sides correspond to different countries. Flows represent production in the countries on the left controlled by the enterprises of the countries on the right. Flow width is proportional to the production in metallic mass content. Colours of flows correspond to country of origin of material production. 'FDI control' measures the proportion of the global mining production controlled by foreign companies. Here, the numerator excludes the output from transnational

corporations in their home countries, while the denominator accounts for all production. The HHI was used to measure the mineral production concentration, as shown at the bottom of each panel (two sides for two perspectives). HHI was calculated as the sum of the squared market shares of each producer, with values ranging from 0 to 10,000. A higher HHI indicates greater market concentration. The US Department of Justice classifies an HHI below 1,500 as competitive, 1,500–2,500 as moderately concentrated and above 2,500 as highly concentrated⁴⁰. ROW, rest of world.

supply concentration (quantified by the HHI using the production and trade data) and the political stability of suppliers. To explore the role of FDI in mitigating supply risk, we derived an 'adjusted SRI' on the basis of the adjusted international trade data (see Methods for details).

Of 240 countries and dependent regions studied, 43% were involved with imports of the four clean energy materials, with 85 countries for lithium, 39 for cobalt, 51 for nickel and 99 for platinum. China's cobalt supply risk was the highest, with an SRI of 88, followed by its nickel and lithium supply with SRIs of 51. Finland's cobalt supply risk, with an SRI of 47, ranks fourth. FDI had a significant impact in reducing supply risks for certain countries. The USA's lithium, United Kingdom's platinum, Japan's nickel, and China's lithium and cobalt saw SRI values drop by 63%, 33%, 21%, 15% and 11%, respectively. This is mainly due to a decline in import dependency for the USA's lithium from 67% to 8%, the United Kingdom's platinum from 100% to 65%, Japan's nickel from

100% to 42%, and China's lithium and cobalt from 83% to 55% and 97% to 73%, respectively. However, the impact of FDI was mixed concerning the sub-indicator 'likelihood of supply disruption'. For example, the likelihood of supply disruption of the United Kingdom's platinum decreased from 0.1 to 0.06, while China's lithium saw an increase from 0.22 to 0.35. Although FDI reduced supply risks for some countries, other nations importing the same commodities experienced heightened risk, creating a 'zero-sum' scenario. SRI values for lithium increased by 10% for 82 countries, cobalt by 8% for 38 countries, nickel by 18% for 50 countries and platinum by 2% for 92 countries after adjustment.

Discussion

Our analysis shows that transnational investment is widely involved in the production activities of lithium, cobalt, nickel and platinum. In the context of economic globalization, large mining companies

Table 1 | Countries that are most affected by FDI on both the supply side and the demand side for the four critical materials

Material	Supply side			Demand side			Global production in 2019 (kt)	Supply concentration (measured by HHI)	
	Country	Domestic production controlled by FDI (kt)	Share of production controlled by FDI in total domestic production (%)	Country	Controlled overseas production (kt)	Reduction of SRI when considering FDI (%)		Geographical perspective	Company ownership perspective
Lithium	Australia	24	54	USA	21	63	87	3,363	2,511
Cobalt	DRC	84	84	China	22	11	144	4,917	1,434
Nickel	Indonesia	323	38	China	287	8	2,507	1,804	1,307
Platinum	South Africa	39 ton	29	United Kingdom	42 ton	33	186 ton	5,348	4,088

All values are in metallic content unit.

from economically advanced countries have a great impact on the supply of critical minerals. Quantitatively, we estimated that 47% of lithium, 71% of cobalt, 41% of nickel and 34% of platinum production were under FDI in 2019.

The market of critical materials is generally deemed to be vulnerable because it is geographically concentrated in a few countries that are not the major consumers, resulting in many countries being highly dependent on exports from limited sources. Taking into account company ownership leads to substantial changes in the roles of many countries in global material production and trade networks. Table 1 summarizes the countries that have been affected the most in each of the four critical materials. On the supply side, lithium in Australia, cobalt in the DRC, nickel in Indonesia and platinum in South Africa are the ones receiving the most overseas capital investment. Therefore, from the company ownership perspective, the influence of the host countries' priorities on production of these materials may not be as great as may be expected from the geographical perspective. Such variations contribute to significant differences in the supply concentration of the four materials between the two perspectives outlined in the HHI, especially for cobalt.

On the demand side, our supply risk assessment results indicate that the USA, the United Kingdom, Japan and China experience an overlap between their overseas investment and domestic demand. If the companies act at least partially in accordance with the interests of material supply security of the countries to which they belong, their overseas investments could, to a certain extent, alleviate the import risks faced by these consuming countries. Our SRI-based quantitative results consider the above scenario, which represents the upper bound of potential impact of FDI. In practical terms, FDI's role should be considerably smaller, given that companies prioritize shareholder profits over resource security of their origin countries. However, the alignment of corporate shareholders' interests with those of consumers in their home country can occasionally occur due to factors such as the integration of downstream capacities in the country of origin, policy incentives based on national security and the interests of state investors.

Here we differentiate between private and state investors and create an alternative scenario for evaluating the role of FDI in mitigating national supply risks. We find that private investors dominate the markets of selected materials, as outlined in Supplementary Table 10. In 2019, the mining production controlled by foreign state investors (including state-owned companies and government capital) was 9.1% for cobalt and 3.2% for nickel, with no involvement in lithium and platinum production. Among state investors, Chinese state-owned companies held the largest shares, controlling 5.2% of cobalt and 2.7% of nickel production in foreign assets. The other shares were mainly held by state investors from Kazakhstan (3.8% of cobalt) and South

Korea (0.5% of nickel). These proportions could be interpreted as a lower limit for the role of FDI in geopolitical risk mitigation. A more precise quantification of the correlation between a company's country affiliation and decision-making priorities could be supported by future empirical analysis at the company level.

Meanwhile, FDI may reduce the availability of raw materials for other countries that do not have the ability to engage in foreign investment. Therefore, communication and cooperation among the multiple stakeholders at both the company and country levels are crucial to avoiding or de-escalating geopolitical tensions that may harm the global green transition.

We also reveal a certain mismatch between the supply controlled by FDI and import demand, especially for cobalt. There are 26 bilateral country pairs with FDI relationships for cobalt, but only 3 bilateral country pairs also have trade relationships. The potential for supply risk mitigation through FDI for cobalt is limited at present.

The extensive prevalence of FDI distribution indicates that, in addition to the domestic stakeholders in the exporting countries (for example, Australia, the DRC, Indonesia and South Africa), international private and state investors from major economies (for example, the USA, the United Kingdom and China) are major players whose actions can affect sustainable material supply. Cross-boundary investment and trade have significantly accelerated the development of mining industries in host countries but also come with many negative social and environmental externalities^{41,42}. These impacts include, for example, water stress and pollution of lithium extraction from brine in Chile and Argentina, human rights issues in artisanal and small-scale mining of cobalt from the DRC and marine pollution issues from high-purity nickel processing by high-pressure acid leaching in Indonesia¹¹. Such challenges could in part be resolved through ongoing and proposed public policies on the environment and human rights; thus, the compliance of domestic and international investors and adequate oversight by governments are indispensable⁴³. The regulatory role of the state in offshore investment needs careful attention, especially for cobalt and nickel as many related mining projects are involved with government equities. Future research could explore the implications of foreign investment in mitigating the environmental and social impacts of local resource extraction, and compare them with the role of indigenous investment.

In the long term, recycling may change or even reverse the roles of resource exporting and importing countries⁴⁴. Nickel and platinum recycling industries are relatively well established, with an end-of-life recycling rate of 60%³⁹ (ratio of recycled material to total material content of end-of-life products), while that of lithium and cobalt is much lower (1% and 15%, respectively)¹⁵. Nevertheless, government policies, together with the improvement of business models and economies of scale brought about by the potential ubiquity of lithium

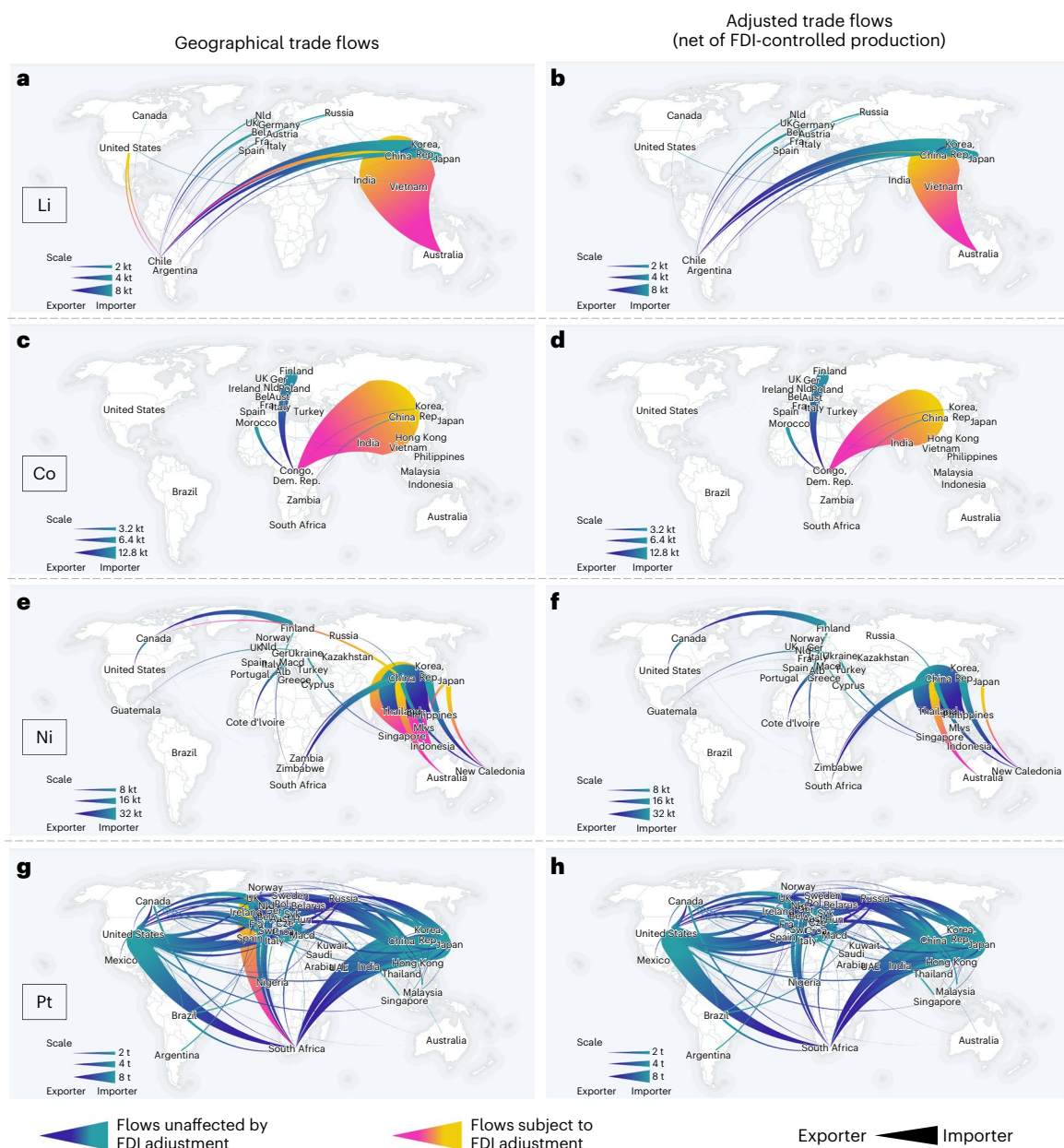


Fig. 3 | Geographical and adjusted trade network of the four clean energy materials in 2019. a–h. The geographical trade flows (left) and adjusted trade flows (right) of (a,b) lithium, (c,d) cobalt, (e,f) nickel and (g,h) platinum. Two coloured flows represent two trade types: one with corporate FDI between partners, subject to adjustment; the other without such investment, remaining unaffected. Adjusted trade flows were derived by subtracting material production controlled by importing country companies in exporting countries

from the geographical trade flows. The flows are directed from the exporting country (narrow end) to the importing country (wide end). Flow width represents the trade amount in metallic content. Due to the large differences in trade value data among different countries, trade flows less than 0.01% of the maximum trade flow are not displayed. Ger, Germany; Nld, the Netherlands; Bel, Belgium; Fra, France; Spa, Spain; Mlys, Malaysia; Alb, Albania; Cze, Czech Republic; Grc, Greece; Svk, Slovakia; Swi, Switzerland; Croa, Croatia.

battery-powered products, are expected to markedly boost the recycling of lithium and cobalt in the future⁴⁵. Technological evolution will contribute to altering global and national critical materials security as well. Clean energy technology development faces many uncertainties, for example, the commercialization of batteries with various innovative chemistry systems⁴⁶ and energy storage solutions for the power sector and long-range heavy-duty transportation⁴⁷. For both investment and policymaking, it is essential to develop long-term strategies and to make timely production adjustments on the basis of evolving technological dynamics.

Various other materials, including copper, rare earth elements and so on, are also identified as critical minerals for clean energy technologies (see Supplementary Table 11). The availability of many of these raw

materials presents a challenge comparable in severity to the materials discussed in this study (for example, the issue of lack of supply diversity⁴). The supply risk assessment model developed in this study can be extended to these critical minerals to evaluate the heterogeneous role of FDI in various markets. Aside from the minor caveats previously described for lithium and platinum, this study only focuses on upstream mining production. FDI also widely exists in the downstream of the industrial chain. The issue of highly concentrated supply is even more pronounced downstream than upstream. Currently, 60%–90% of the production capacity of intermediates, components, final products and recycled scraps are each dominated by top three producers^{21,48}. The perspective and methods here could be applied across the entire industrial chain of clean energy technologies, although several challenges

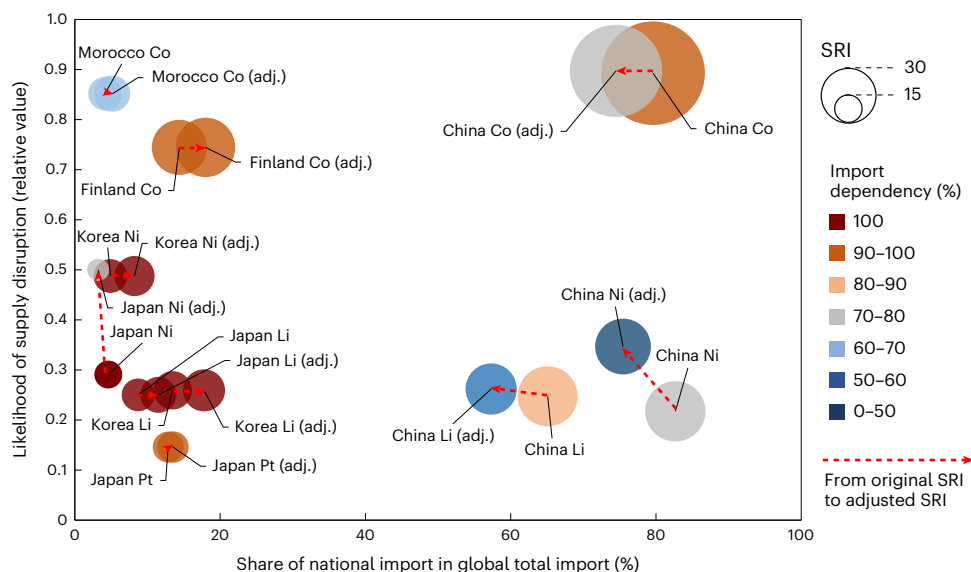


Fig. 4 | Top 10 supply risk indices. Circle size represents the value of SRI. The horizontal (share of national imports in total global import) and vertical (likelihood of supply disruption) axes represent two of the three sub-indicators used to calculate the SRI. The colour of the circle represents the third sub-indicator used to calculate the SRI: import dependency. The likelihood of supply

disruption was derived from the supply concentration (measured using HHI) and the political stability of suppliers, which represents relative rather than actual probabilities in the interval 0–1. ‘Adj.’ stands for ‘adjusted’ SRI that considers the impact of FDI.

would need to be overcome. Challenges include collecting production and trade data from both the geographical and company ownership perspectives, establishing a comprehensive data-processing framework to transform and integrate raw data from various sources, and incorporating new functional modules to evaluate the vulnerability of energy technologies to supply restrictions at different stages.

Methods

System boundary

We chose four critical materials for clean energy transition in this Analysis: lithium, cobalt, nickel and platinum. We focused on the FDI related to primary supply, that is, the mining projects. To ensure that the assessment results are comparable across the different materials, we consistently considered the mining stage for all four materials when mapping the production flow. For lithium brines and platinum ores mining projects, these are often vertically integrated with refining facilities. This integration occurs because transporting these mineral products for international trade is generally unprofitable. It follows that our system coverage for international trade included both the mining and refining stages for lithium and platinum, and only the mining stage for cobalt and nickel. Therefore, under our framework, the selected commodity types encompassed ores, brines and lithium carbonates for lithium; ores for cobalt; ores for nickel; and ores and refined metals for platinum, as illustrated in Supplementary Fig. 13. These commodities are collectively referred to as ‘materials’ for reading convenience.

This study covered 193 sovereign states and their dependent territories (240 regions in total). The country classification was based on ISO 3166 (ref. 49). The temporal boundary of material production and international trade data was set to the year 2019 as it was the most recent year with sufficient data. The information about company FDI and ownership hierarchy was updated to the end of the first quarter of 2020.

Tracing material production and international trade

Production data for the four materials were adopted from the United States Geological Survey^{14,50–53}. International trade data for material-related commodities were adopted from the United Nations

Comtrade database and Customs databases of major countries, using the custom codes of corresponding commodities^{54–57}. Material contents in relevant commodities were preliminarily determined on the basis of commodity industrial properties and amended on the basis of mass balance principles, which require the sum of national domestic production and imports to be approximately equal to the sum of exports and domestic consumption. We employed two main methods for balancing each country’s inputs and outputs. The first method adjusts traded commodity values on the basis of production and consumption reports, aligning metal content with net trade differences, applicable in countries with detailed mineral reporting. The second method estimates metal unit production for countries lacking specific data, using physical tonnage and referencing trade values from major producers. The reliability of the database was verified by a series of related material flow analysis studies previously published by the authors^{8,19,36–38}.

Material production control quantification

Company ‘control’ is typically defined as the ability to influence substantial strategic decisions regarding operations and investments. Referencing the practices in existing literature³⁰, we quantified control using the companies’ ownership of mine projects and subsidiaries. We assumed that control is proportionate to each shareholder’s percentage of ownership. Then the control over material production could be attributed to the relevant companies on the basis of their equity ownership share structure. This approach was selected due to its capacity to provide a clear, visual representation of all stakeholders.

We established a critical material FDI database that collects the hierarchy of mine producers, relevant holding companies, intermediary companies and upstream parent companies. The database contains 32 lithium mines, 42 cobalt mines, 49 nickel mines and 31 platinum mines worldwide that are involved with FDI activities (note that lithium brine and platinum ore projects also tend to consolidate refining capacities). The information about mine production, owners and shareholders of these mines came from annual reports, public statements and news reports of relevant companies, consulting groups and information services platforms. The information about quantitative equity ownership relationships between companies and

country affiliation of companies (defined by headquarter locations) was extracted from the ORBIS database of Bureau van Dijk, a professional analytics company⁵⁸. The ORBIS database is a comprehensive global resource that offers detailed financial information on 45 million companies worldwide, including their trade descriptions, extensive corporate ownership structures, mergers and acquisitions deals and rumours, which are captured from hundreds of separate providers and Bureau van Dijk's own sources.

Here we only considered the companies that engaged in the production activities of the material-related industry chain, such as mining, chemical processing and downstream manufacture. The ownership of companies that are involved in investment capital in mines but not in the specific operating activities was classified to the country where the mine is located because these investment companies have relatively limited influence over the mine production activities. Our results were verified in comparison with those of relevant existing literatures^{30,31,59}. Detailed data are transparently shown in Supplementary Figs. 1–4 and Tables 1–4.

Supply risk index calculation

Supply risk could be triggered by several factors, including accidental hazards of mining activities or the natural environment of the location (for example, natural disasters, epidemiological outbreaks), and intentional accidents caused by local communities or governments (for example, trade restrictions, strike procession). In this study, we focused on geopolitical supply risk, which we quantified using the 'SRI' indicator that we developed. As the country's influence over domestic supply against geopolitical factors is much stronger than its control over foreign supply, we considered domestic production as a 'risk-free' source of supply, although this assumption may not always align with reality⁶⁰. Built upon existing literatures^{61,62}, the primary components of import risk that we considered encompassed national import dependence, supply concentration and political stability of foreign suppliers. We further modified the import risk assessment framework in these literatures to focus more on the national import structure. On the basis of these considerations, we calculated the SRI across three dimensions: (1) import dependency, (2) likelihood of supply disruption and (3) share of national import in total global import.

Import dependency measures the degree of dependence of a country on foreign supplies, as shown in equation (1). The higher the dependence on imports, the more unstable the supply structure of the country.

$$ID_i = \frac{TIM_i}{TIM_i + PO_i} \quad (1)$$

where ID_i is the import dependency of country i , TIM_i is the total import amount of country i , and PO_i is the domestic production of country i .

Likelihood of supply disruption measures the relative likelihood of disruption under different import combinations and was calculated using equation (2). This indicator was derived on the basis of the HHI and the regional political stability index. HHI is the sum of the square of the production shares of each producer⁶³. A high HHI value represents high supply concentration, that is, a large likelihood of supply disruption. Further, we used the World Bank's Worldwide Governance Indicator (WGI) to reflect the impact of national-level government management on supply risk⁶⁴. The WGI contains six measurement dimensions; we used the one called 'political stability and absence of violence/terrorism' for the year 2019 and converted it to the interval from 0 to 1, where a high value of the indicator reflects a low governance level.

$$LSD_i = \sum_j \left(\frac{IM_{i,j}}{TIM_i} \right)^2 \times WGI_j \quad (2)$$

where LSD_i is the likelihood of supply disruption, $IM_{i,j}$ is the import of country i from country j , and WGI_j is the worldwide governance index (dimension of political stability and absence of violence/terrorism) of country j .

Other indices fulfil a role similar to that of the WGI, for example, the Policy Perception Index (PPI) developed by the Fraser Institute⁶⁵. To examine the impact of varying indicator selections on the uncertainty of our results, we conducted identical calculations for SRI using the WGI and PPI. A comparison of the findings indicates that while there are discrepancies in the quantitative outcomes, the qualitative conclusions of this study largely remain consistent. Detailed information on these two indicators, along with a comparison of their results, can be found in Supplementary Fig. 14 and the accompanying explanatory text.

The share of national import in total global import measures the relative size of a country's total demand, as calculated following equation (3). A country with a larger share indicates that the country has more difficulty than other countries in adjusting its supply to meet the import demand¹⁵.

$$IS_i = \frac{TIM_i}{GIM} = \frac{TIM_i}{\sum_i TIM_i} \quad (3)$$

where IS_i is the share of the import of country i in the total global imports, and GIM is the total global imports.

The three aforementioned indicators were designed to encompass the three distinct dimensions of risk: (1) exposure, represented by import dependency; (2) hazard, represented by the likelihood of supply disruption and (3) vulnerability, represented by the share of national import in total global import. We adopted the methodology from a previous study⁶² to integrate the three indicators into an overall risk index, SRI, by multiplying them and further normalizing them to a range of 0 to 100, using equation (4).

$$SRI_i = (ID_i \times IS_i \times LSD_i)^{\frac{1}{3}} \quad (4)$$

where SRI_i is the supply risk index of country i .

To explore the impact of FDI on supply risk mitigation, we further proposed the concept of 'adjusted supply index'. In the evaluation of this index, we treated the production through FDI as a risk-free source of supply, similar to domestic production. The computational structure of the index is unchanged, that is, still based on a combination of sub-indicators in three dimensions. For the calculation of each sub-indicator detailed above, we subtracted the capacity controlled by FDI from the corresponding import amount and added it to the domestic production, as equations (5)–(8) show.

$$SRI_{adjusted,i} = (ID_{adjusted,i} \times IS_{adjusted,i} \times LSD_{adjusted,i})^{\frac{1}{3}} \quad (5)$$

$$ID_{adjusted,i} = \frac{TIM_i - \sum_j P_{i,j}}{TIM_i - \sum_j P_{i,j} + PO_i + \sum_j P_{i,j}} = \frac{TIM_i - \sum_j P_{i,j}}{TIM_i + PO_i} \quad (6)$$

$$LSD_{adjusted,i} = \sum_j \frac{(IM_{i,j} - P_{i,j})^2 \times WGI_j}{(TIM_i - \sum_j P_{i,j})^2} \quad (7)$$

$$IS_{adjusted,i} = \frac{TIM_i}{GIM} = \frac{TIM_i - \sum_j P_{i,j}}{\sum_i (TIM_i - \sum_j P_{i,j})} \quad (8)$$

where $P_{i,j}$ is the production controlled by country i through FDI in country j .

Limitations and uncertainties

Statistical biases and propagation errors in the underlying data lead to uncertainties in our results. We deemed data sources of geographical production and trade of raw materials to be relatively robust since they have been verified by a series of previously published research^{15,19,36–38}. Similarly, the potential impact of the inaccuracies of data on property rights of mining enterprises due to unrecorded investments is limited because the ownership information has been verified by the authority

database of Raw Materials Data³⁰. While the potential flaws in model structure and parameters (for example, WGI) may make the quantitative results of the model partially inconsistent with reality, the qualitative conclusions still hold as verified by the results of calculations using other parameters with the same functions (Supplementary Fig. 14).

Our method of quantifying company control via ownership only might have overlooked nuanced factors such as discrepancy of government and private shareholders, country-specific government–business connections, subjective tendencies of decision makers and company-specific decision-making processes. Further studies are needed to accurately quantify the influence of investment and production decisions, considering the characteristics of the various stakeholders and their complex interaction processes. Compared with some studies that reassigned total control to companies surpassing certain ownership thresholds (for example, 20%) or to the highest ownership stakeholder³⁰, our approach might have overestimated minor shareholders' control while underestimating larger shareholders' influence in some cases. The uncertainties and suitability of these varying methodologies need to be examined in future studies.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The information on country affiliation of companies is from the ORBIS database of Bureau van Dijk⁵⁸. The ownership hierarchy of mine owners and stakeholders was collected from annual reports, public statements and news reports of relevant companies, consulting groups and information services platforms. Geographical production and international trade of four materials are from the United Nations Comtrade database⁵⁴, United States Geological Survey Reports¹⁴ and our previous studies^{8,19,36,37,66}. Source data are available in Supplementary Information.

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Author contributions

X.S. and H.H. conceived the idea, conducted the analysis and wrote the paper. C.G., T.F. and G.L. wrote sections of the paper and provided guidance on the paper. D.X. and M.E. provided data for the analysis. I-Y.L.H., Z.L. and F.Z. reviewed and edited the paper. H.H. and G.L. supervised the work and secured funding for the project.

Competing interests

The authors declare no competing interests.

Additional information

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Sampling strategy	No sample-size calculation was performed because this study is an observational study of global lithium, cobalt, nickel, and platinum mining industry.
Data collection	The data was collected from publicly available source by direct download or copying. The data collection was conducted by the authors of this study without using specific software or coding work.
Timing and spatial scale	The data was mainly collected from January 2020 to June 2020 and revised and reverified from June 2023 to September 2023.
Data exclusions	No data were excluded from the analyses.
Reproducibility	The results of this study can be easily reproduced through the steps described in the Methods section, using the source data provided in the Supplementary Information. The analysis could be conducted using Microsoft excel or similar software.
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